Lab Experiment 9

Travelling Salesman Problem (TSP)

Aim

To write a Python program to solve the Travelling Salesman Problem (TSP).

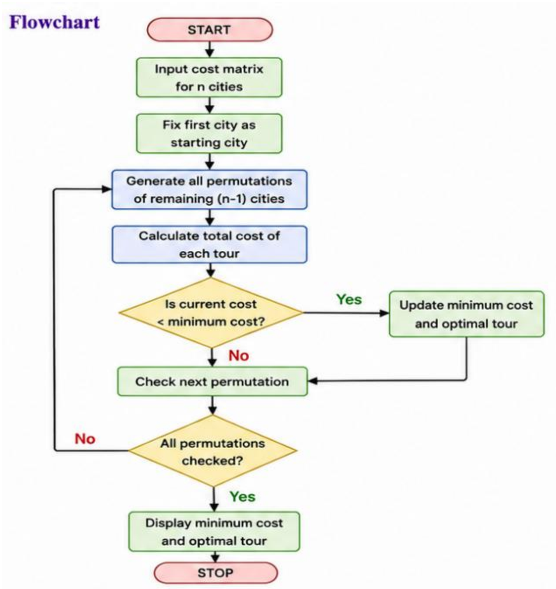
Objective

The objective of this experiment is to implement the Travelling Salesman Problem using Python. It aims to find the shortest possible route that visits every city exactly once and returns to the starting city.

Algorithm

1. Define the cost matrix representing the distances between cities.
2. Generate all possible routes starting from the first city.
3. Calculate the total cost of each route.
4. Compare the costs and keep track of the minimum cost.
5. Store the corresponding optimal route.
6. Display the minimum cost and the optimal tour.

Flowchart



Code

```
from itertools import permutations


graph = [
    [0, 10, 15, 20],
    [10, 0, 35, 25],
    [15, 35, 0, 30],
    [20, 25, 30, 0]
]


n = len(graph)
cities = list(range(1, n))


min_cost = float('inf')
best_path = []
```

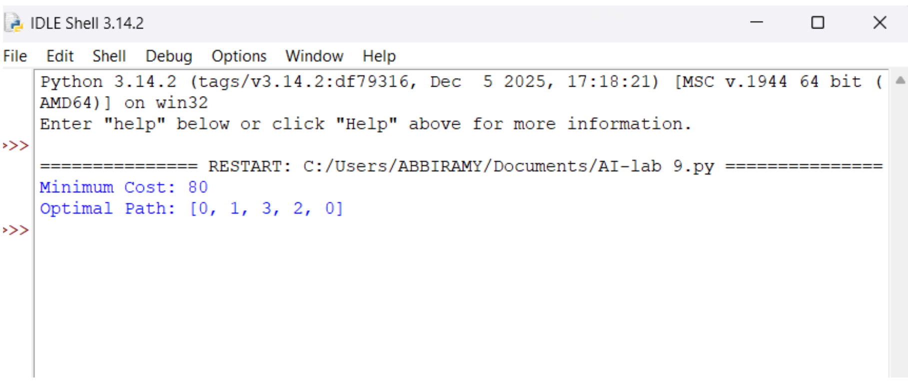
```
for path in permutations(cities):
    current_path = [0] + list(path) + [0]
    cost = 0

    for i in range(len(current_path) - 1):
        cost += graph[current_path[i]][current_path[i + 1]]

    if cost < min_cost:
        min_cost = cost
        best_path = current_path

print("Minimum Cost:", min_cost)
print("Optimal Path:", best_path)
```

Output



Result

The Python program successfully found the minimum travelling cost and the optimal path for the given Travelling Salesman Problem.

Conclusion

The Travelling Salesman Problem helps determine the shortest possible route that visits all cities exactly once and returns to the starting city. The brute-force approach guarantees the optimal solution for small datasets.